

- 4 The area of a triangle is 12 cm^2 .
The length of the base of the triangle is 8 cm .

Work out the height of the triangle.

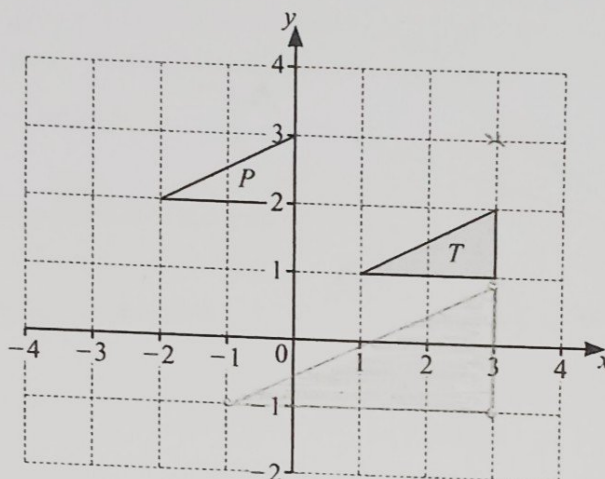
$$= \frac{1}{2} \times b \times h \quad h = 3 \text{ cm}$$

$$12 = \frac{1}{2} \times 8 \times h$$

$$\frac{24}{8} = \frac{8h}{8}$$

5

..... 3 cm [2]



$$A = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$$

$$C = \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$$

- (a) Describe fully the **single** transformation that maps triangle T onto triangle P .

Translate triangle T by the vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$

- (b) Draw the image of triangle T after an enlargement of scale factor 2, centre $(3, 3)$.

- 6 Find the value of

(a) $5^{-5} \times 5^5$

(b) $125^{\frac{2}{3}}$

$$\left(\sqrt[3]{125} \right)^2 = (5)^2 = 25$$

..... 1 [1]

..... 25 [2]